

Data cleaning and Filtering

- Missing Data
- Erroneous values
- Type ~~conversion~~ Conversion
- ~~Final~~ Entity resolution
- Data integration

Start with unimodal summary
then start to consider relationships
among variables

Data Adjustment: Relational Algebra

- Relational data model
- Data Transformation (SQL)
- Projection (select) - select columns
- Selection (where) - filter rows
- Sorting (order by)
- Aggregation (group by, sum, min, max)
- Combine relations (union, join)

Idk adjust it

- Roll up $\rightarrow$ smaller
- Drill down $\rightarrow$ bigger

Visual language

- Visual language is a sign system
 - may perceive as a set of signs
 - Sender encodes information in signs
 - Receiver decodes information from signs

How and when



value encoded $\rightarrow$ ordinal variable



Encodes continuous variable
(10)